

Mateus Pereira da Silva

Senior Product Engineer | Full Stack · AI Systems · Distributed Infrastructure

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SUMMARY

Senior Product Engineer with **5+ years** shipping software from ambiguous problem to production across product, backend, frontend, data, infrastructure, and AI. Owns product and AI areas end to end at VE3 on a multi-tenant B2B SaaS platform used by **30+ enterprise customers**, with three promotions in four years. Builds production AI workflows, distributed services, data systems, and customer-facing products, then stays responsible for deployment, observability, cost, latency, failures, and incident response. Outside work, co-founded a zero-to-one travel product and builds open-source AI and infrastructure tooling.

TECHNICAL SKILLS

Languages & frontend: Go, TypeScript, Python, JavaScript, SQL, PHP; React, Next.js, Vite, React Flow, Tailwind CSS

Backend & systems: REST APIs, FastAPI, Flask, microservices, distributed systems, SDKs, background workers, event-driven architecture, workflow orchestration

AI: RAG systems, agent workflows and orchestration, tool calling, structured outputs, context management, prompt engineering, model selection, embeddings, AI evaluation, MCP, code agents; LangGraph, LangSmith, Claude API, OpenAI API

Data: PostgreSQL, MySQL, Redis, Apache Kafka, Prisma, Neon Postgres, Polars, data modeling, query optimization, data pipelines

Cloud, operations & quality: AWS (Lambda, SQS, ECS, S3, RDS, API Gateway), Docker, Kubernetes, Terraform, Cloudflare, Linux, CI/CD; Prometheus, Grafana, CloudWatch, pytest, Jest, go test, Playwright, Selenium

Architecture, security & product: System design, multi-tenant systems, scalability, technical trade-offs, threat modeling, hardening, access and secrets; product scoping, discovery, MVP definition, prototyping, stakeholder alignment, UX thinking

PROFESSIONAL EXPERIENCE

VE3 | Enterprise Data Platform (B2B SaaS)

Senior Product Engineer

Aug 2022 - Present | Remote

Promotion path: Intern > Junior > Mid-level > Senior. Promoted three times in four years.

- Own product and AI areas from problem discovery and data modeling through implementation, deployment, monitoring, and production support; partner directly with the CEO, stakeholders, product, design, users, and customer API integration teams.
- Lead technical design for platform and AI features and initiate internal projects outside the roadmap when there is clear product or engineering leverage; support teammates through design reviews and knowledge sharing.
- Shipped a production RAG system using LangGraph across backend and frontend, then used LangSmith traces and product usage data to evolve it into an agent experience with an orchestrator and specialist agents that use tools inside the product.
- Own model selection, prompt design, context management, latency, failure handling, and cost for production AI systems.
- Built REST APIs, SDKs, background workers, and Go/Python microservice orchestration for high-volume, multi-tenant workflows.
- Built Python/FastAPI and TypeScript services and event-driven AWS workloads using SQS, Lambda, ECS, RDS, S3, and API Gateway; provision infrastructure with Terraform alongside Docker-based services.
- Built the distributed infrastructure behind a forecasting platform running **25+ predictive models** over millions of records.
- Improved the platform's highest-traffic customer workflow by **75%** through PostgreSQL optimization and reduced dashboard query cost by roughly **50x** in another heavily used area.
- Established observability with Prometheus, Grafana, and CloudWatch; maintained approximately **three years without unplanned production downtime** while owning deployments and operational response. Use pytest and Playwright for automated testing.
- Use AI-assisted development daily for research, debugging, implementation, test generation, and architecture exploration, reviewing and validating outputs before production use.

Self-Employed

Software Developer

Feb 2021 - Aug 2022 | Remote

- Built and shipped websites, web applications, APIs, databases, and automation systems for small businesses and clients, owning requirements, implementation, deployment, and client communication.
- Translated loosely defined business needs into production software that removed repetitive operational work.

SELECTED PROJECTS

Workflow Execution Engine (WEE) | Founder & Engineer

Go · JSON Schema · CLI · LLM APIs · React · React Flow

Open-source runtime for controllable AI workflows. Built and validated a **21-step** engineering workflow with **7 model workers** and **14 deterministic steps**; the engine owns retries, branching, replay, schema-enforced outputs, observability, context policies, and execution budgets, and ships as a single static Go binary.

Plan Patagonia | Co-founder & Product Engineer (0 to 1)

TypeScript · Next.js · React · Prisma · Neon Postgres · Docker · S3 · Cloudflare · Claude API

Building a zero-to-one travel platform that adapts Patagonia itineraries using weather, reservation, and transportation data. Own product direction, traveler experience, architecture, AI integrations, frontend, backend, infrastructure, roadmap, and delivery.

Kafka Go SDK | Open Source

Go · Apache Kafka

Reusable Kafka infrastructure for connection handling, retries, producers, consumers, and error management so services can focus on shipping events instead of rebuilding plumbing.

LANGUAGES

Portuguese (Native) | English (Advanced, C1) | Spanish (Beginner)